Bayesian networks – assignment part 2

The goal of this part of the assignment is to complement your implementation of the Variable Elimination algorithm with MAP queries (computing the most probable explanation of observed variables) and apply it on MAP queries in different Bayesian networks. Keep a log (textfile) of the steps your algorithm does on a particular network with a particular elimination ordering according to the steps on the slides, e.g., what are the query/evidence variables, what are the factors, what is the elimination ordering, which factors are processed, etc. and extend this now with the added steps for MAP inferences.

There is code available in Java and Python for reading in a network from a file, and some structure to help you focus on the core of this assignment: efficiently implementing the algorithm using suitable data structures. The Bayesian Network repository (http://www.bnlearn.com/bnrepository/) gives you many examples of real-world (and less-real-world, but known benchmarks) networks. You can also construct example networks yourself. The example code uses the .bif format. We also attach a .bif of the Endorisk model. Note some networks are unlikely the be able to process complex MAP queries (with 5 or more query variables) due to huge factor sizes.

This part of the programming assignment also comes in two parts:

- In **1a** we asked you to implement multi-dimensional factors and do the necessary calculations on them (reduction, product, marginalization)
- In **1b** you've built on this to implement variable elimination for inference.
- In **2a** you will implement the maximalization calculation on factors (if not already done)
- In **2b** you will complement the variable elimination algorithm with MAP queries

The adjusted Excel sheet for part 2 will give you an indication of the weight of various components of the assessment. **An extra bonus point** if offered if you explore the limits of tractability and show in your report when MAP breaks down due to memory shortage and/or taking too much time.

To see if your algorithm's results are correct, you can compute the joint probability distribution over the MAP variables and find the value assignment with the highest probability. Note that this is a highly inefficient way to implement MAP and this is for sanity checks only!

Incorporate formative feedback

Please incorporate the formative feedback on the first part of the assignment in the remainder of the assignment. This feedback should give you sufficient pointers for improvement where needed for a passing grade. For the final submission (parts 1, 2, and 3 combined), please indicate clearly how and where you made improvements based on the formative feedback. For several parts (e.g. readability, testing) an overall grade is given – this can be lower than the formative feedback when the overall quality decreases!